MAHATMA GANDHI UNIVERSITY
FOURTH SEMESTER UG PRACTICAL EXAMINATION
MG4DSCCHE202- Fundamentals of Physical Chemistry- Practical

Model Question Paper

Time **-3 hr.** Total marks: **35**

Part-I (15 marks)

Conduct **one** Experiment with √ mark (**Select One**)

1. Determine the transition temperature T1 of Sodium thiosulfate pentahydrate using a cooling curve for a saturated solution in equilibrium with excess solid.
2. Determine the critical solution temperature (CST) and corresponding composition for the partially miscible phenol-water system.
3. Determine the Equivalent point and strength of the given HCl solution by titrating it conductometrically against standard NaOH solution.
4. Determine the molecular mass of a non-volatile Organic compound (acetanilide or p-dichlorobenzene) using the Rast method based on the freezing point depression of Naphthalene.
5. Determine the equivalent point and concentration of Fe2_+ ions by potentiometric titration with standard potassium dichromate solution in an acidic medium**.**
6. To construct the temperature – composition phase diagram for a binary organic system like naphthalene-biphenyl, and obtain the temperature and eutectic composition from cooling curves.
7. Determine the Viscosity of a given sucrose solution and find its percentage composition by comparing it with the viscosities of standard sucrose solutions of known concentrations.

Part-II (5 marks)

8**.** Write the principle and Procedure for the given Experiment **(Select One from the above list of 7)**

9. Viva voce based on the practical **(10 Marks)**

10. **Lab report** **(5 marks)**

Instructions

1. Give one experiment from Part-I.

2. Select one questions from part I to write the principle and procedure for **Part II.**